

Brain Tumor Classification from MRI Images Using Deep Learning

1. Abstract

This project focuses on classifying brain tumor types from MRI images using deep learning models. The task is a multi-class classification problem, where each image is assigned to one of four categories: glioma, meningioma, pituitary tumor, or no tumor. This problem is important because accurate tumor classification can help support medical diagnosis and reduce the workload for healthcare professionals.

We use a public MRI dataset from kaggle and compare two types of models: a baseline convolutional neural network (CNN) and a more advanced transfer learning model based on EfficientNet. The baseline CNN is a relatively simple architecture trained from scratch, while the EfficientNet model uses pretrained weights from ImageNet and is further fine-tuned on the MRI dataset.

To evaluate the models, we use several metrics including accuracy, precision, recall, F1-score, and ROC-AUC. The baseline CNN achieves around 70% accuracy, but shows weaker performance on some classes, especially meningioma. In contrast, the EfficientNet model improves the overall accuracy to about 85% and provides more balanced results across all classes. It also achieves higher AUC values, indicating better classification ability.

Overall, the results show that transfer learning significantly improves performance compared to a simple CNN. Using pretrained features, along with data augmentation and fine-tuning, allows the model to better capture patterns in MRI images. However, some challenges remain, especially when different tumor types look visually similar or when image quality is low.

To address the issue with medical diagnostic errors due to similarity, we implemented a cost-sensitive learning strategy using a weighted Categorical Cross-Entropy loss function (weights: 1.5, 2.2, 1.0, 0.6). This adjustment specifically targeted the lower precision and recall observed in the Glioma and Meningioma classes. Our final Advanced Model, leveraging the EfficientNet-B0 architecture with a ReduceLROnPlateau learning rate scheduler, demonstrated superior convergence stability. The results indicate that while baseline CNNs provide a functional starting point, the combination of compound scaling and transfer learning is essential for achieving the sensitivity required for clinical decision support systems.

In addition to the accuracy improvement, the baseline CNN, while functional, demonstrated a critical failure in sensitivity, particularly with a **0.40 Recall** for the Meningioma class. In contrast, the **EfficientNet-B0** model—augmented by a **weighted loss vector** and a **ReduceLROnPlateau scheduler**—attained a macro-average ROC-AUC of 0.96. This represents a **30% increase in recall** for the most difficult tumor types, validating that the combination of compound scaling and cost-sensitive learning is essential for clinical-grade diagnostic reliability.

2. Research Objective and Classification Task

2.1 Introduction

In this project, we study the problem of brain tumor classification using MRI images. This is a classification task where the goal is to assign each image to one of four categories: glioma, meningioma,

pituitary tumor, or no tumor. Compared to traditional machine learning methods, deep learning models can automatically learn features from images, which makes them suitable for this type of problem.

We build and compare two models. The first one is a baseline CNN model, which is a standard architecture with convolution and pooling layers. The second one is a more advanced model based on EfficientNet, which uses transfer learning. The idea is that the model has already learned useful visual features from a large dataset, and we can adapt those features to our MRI classification task.

2.2 Motivation

The clinical significance of automated brain tumor identification cannot be overstated. Brain tumors are among the most lethal forms of cancer, and successful treatment is heavily predicated on early and precise diagnosis.

- **Reduction of Diagnostic Latency:** In traditional clinical workflows, radiologists must manually review hundreds of MRI slices per patient. An automated system acts as a "second pair of eyes," flagging anomalies and reducing the time-to-treatment.
- **Mitigation of Human Error:** Inter-observer variability—where two experts may disagree on a subtle scan—is a known challenge in oncology. Deep learning models provide a consistent, objective baseline for classification.
- **Resource Optimization:** In regions with a shortage of specialized neuro-radiologists, automated tools can support general practitioners in triaging cases, ensuring that the most critical patients receive specialist attention first..

However, this task is not easy. Different tumor types can look very similar in MRI images, and sometimes the image quality is not very good. This makes it difficult for models to distinguish between classes. Because of these challenges, it is important to explore more advanced models and techniques, such as transfer learning.

2.3 Objectives and Evaluation Metrics

The main objective of this project is to compare the performance of a simple CNN model and a transfer learning model for brain tumor classification. Instead of only focusing on accuracy, we also consider other evaluation metrics such as precision, recall, F1-score, and ROC-AUC.

Accuracy gives an overall measure of correctness, but it may not reflect performance on individual classes. Precision and recall help us understand how well the model performs for each category, especially for harder classes. F1-score provides a balance between precision and recall. ROC-AUC is also included to measure how well the model separates different classes.

By using multiple metrics, we can better evaluate whether the model performs consistently across all tumor types, instead of only doing well on easier cases.

3.Related Work

In recent years, deep learning has become one of the most widely used approaches for brain tumor detection using medical imaging data. Traditional methods usually relied on hand-crafted features such as texture descriptors, shape information, or statistical features extracted from MRI images. Although these methods can work in some simple cases, they often require professional knowledge in the medical area and are not flexible enough when dealing with complex cases or in high-dimensional data. Reality problems are always hard to solve. As a result, their performance is usually limited when compared to

modern deep learning approaches.

With the development of convolutional neural networks (CNNs), many of these limitations have been addressed. CNNs are able to automatically learn hierarchical features directly from raw image data, which removes the need for manual feature engineering and improves this kind of problem efficiently. According to Kumar et al. (2021), CNN-based models have been widely adopted in brain tumor detection and generally achieve better performance than traditional machine learning methods. These models learn low-level features such as edges and textures in early layers, and more complex patterns in deeper layers. This leads to a higher productivity for medical image classification. However, there is a problem that CNN models often require large amounts of labeled data. In medical applications, datasets are usually relatively small, which can lead to overfitting when training a model from scratch.

To overcome this limitation, transfer learning has been introduced as an effective strategy. Instead of training a model from the initial part, transfer learning uses models that have already been trained on large enough datasets such as ImageNet. These pretrained models can then be fine-tuned for a specific task. Shaha and Pawar (2018) think that using pretrained architectures such as AlexNet, VGG16, and VGG19 can significantly improve classification performance. In particular, more advanced models like VGG19 are able to capture more complex and robust features, which leads to a higher accuracy in image classification tasks. Another advantage of transfer learning is that it reduces training time and improves model stability when the available dataset is limited. This is what we want in a medical problem.

More recent research has focused on improving CNN architectures themselves. One representative example is EfficientNet, which introduces a compound scaling method to balance network depth, width, and input resolution. Instead of arbitrarily increasing one dimension of the model, EfficientNet scales all dimensions in a structured way, leading to better performance with fewer parameters. Verma and Bohat (2025) apply EfficientNet to brain tumor detection and report very high classification accuracy, close to 98%. Their study also emphasizes the importance of data augmentation techniques such as rotation, flipping, and scaling, which help increase dataset diversity and improve generalization. In addition, they show that combining EfficientNet with transfer learning can further enhance model performance, especially in medical imaging tasks where data is limited.

Although existing studies report strong performance, many of them mainly focus on achieving high accuracy using a single model. There is often less emphasis on directly comparing different types of models under the same experimental setting. In particular, it is not always clear how much improvement comes from using more advanced architectures compared to simpler baseline models.

In this project, we try to address this by using comparison between a standard CNN model and an EfficientNet-based transfer learning model. Both models are trained and evaluated on the same dataset using the same evaluation metrics, such as accuracy, precision, recall, and F1-score. By keeping the experimental setup the same, we are able to better understand the actual influence of model choice. This allows us to evaluate whether increased complexity of EfficientNet leads to beneficial performance improvements, and whether transfer learning provides a strong advantage over the traditional CNN models.

4. Dataset and Exploratory Data Analysis (EDA)

4.1 Dataset Overview

The dataset utilized for this study is the Brain Tumor MRI Dataset, sourced from Kaggle. This collection is a combination of the figshare, SARTAJ, and Br35H datasets, curated to provide a robust foundation for multi-class medical image classification.

- Task Type: 4-class classification (glioma, meningioma, pituitary, and no tumor).
- Total Samples: The dataset consists of 7,021 MRI images in total.
- Data Splitting Strategy: Training Set: 4,570 images (approximately 80% of the original training folder).
 - Validation Set: 1,140 images (approximately 20% of the original training folder, used for hyperparameter tuning).
 - Test Set: 1,311 images (held out entirely to evaluate final model generalization).
 - Feature Dimensions: All input images are RGB with dimensions of $224 \times 224 \times 3$ pixels

4.2 Exploratory Data Analysis (EDA)

A critical component of our EDA was verifying the distribution of classes to ensure the model would not develop a bias toward a specific tumor type.

4.2.1 Class Distribution

Analysis of the training data revealed a perfectly balanced distribution across the four categories.

- Glioma: 25.00%.
- Meningioma: 25.00%.
- No Tumor: 25.00%.
- Pituitary: 25.00%.

This balanced nature is ideal for classification tasks as it allows us to use standard Accuracy as a reliable primary metric, alongside Precision, Recall, and F1-score to monitor class-specific performance.

4.2.2 Visual Inspection

Initial visualization of sample images from the dataset highlighted significant challenges.

- Inter-class Similarity: We observed that glioma and meningioma scans often share similar intensities and shapes, which suggested potential "failure modes" during model training.
- Contrast Variability: The MRI scans exhibit varying levels of brightness and contrast, necessitating a robust normalization and augmentation strategy.

4.3 Preprocessing and Data Augmentation

4.3.1 Data Cleaning and Normalization

To prepare the raw MRI data for the convolutional neural networks (CNNs), we implemented the following steps:

- Rescaling: Pixel values were normalized from the range $[0, 255]$ to $[0, 1]$ by multiplying by a factor of $1/255$. This ensures faster convergence during gradient descent.
- Target Resizing: All images were consistently resized to 224×224 to match the input requirements of the EfficientNetB0 architecture.

4.3.2 Explicit Data Augmentation Strategies

Given that the dataset size is relatively limited for deep learning (approx. 7k images), we employed an

aggressive data augmentation strategy using the ImageDataGenerator class to artificially expand the training variety

Strategy	Parameter/Range	Purpose
Rotation	20 degree	Account for variations in patient head positioning during the MRI scan
Width/Height Shift	0.2	Simulate slight off-center alignments in the imaging frame
Zoom Range	0.2	Handle variations in the scale or distance of the tumor within the scan
Brightness	0.7 - 1.3	Normalize the model against different MRI machine settings and lighting conditions
Horizontal Flip	True	Double the dataset by reflecting images across the vertical axis, as brain symmetry is a key feature

Why these strategies were chosen: These augmentations were selected to improve model generalization and reduce overfitting. By exposing the model to "noisy" versions of the data, we force the filters to learn essential pathological features rather than memorizing specific pixel orientations

5. Methodology

5.1 Baseline Model Implementation

We first implemented a standard Convolutional Neural Network(CNN) to establish a performance floor. The architecture follows a hierarchical structure designed to capture spatial dependencies. Our model utilizes a sequence of three "Convolutional Blocks," where each block is responsible for increasing the abstraction of the input data. Detailed of the exact sequences looks like:

Input $\rightarrow$ [*Conv* $\rightarrow$ *ReLU* $\rightarrow$ *MaxPool*] * 3 $\rightarrow$ *Flatten* $\rightarrow$ *Dense* $\rightarrow$ *Softmax*

For input layers, we accept a 224 x 224 grayscale or RGB MRI slice, representing the raw pixel intensities of the brain tissue. The reason to use 224 is because we want to have higher resolution so that the model can notice the small details of the brain tumor images.

In each of these layers, we use the standard operation of 3 x 3 kernels with ReLu activation. We decided to use this small receptive field to capture fine-grained textures (like the margins of a tumor) while keeping the parameter count manageable to prevent overfitting on our specific dataset. The equation looks

like the following:

$$y = \text{ReLU}(W * x + b)$$

W represents the 3x3 kernels and x is the input brain MRI slice. This model was used to evaluate how well basic feature extraction (edges, textures) could identify tumorous vs. non-tumorous regions without transfer learning.

Then following each layer, a Max-Pooling layer with a 2 x 2 window is applied. This reduces the spatial dimensions by half, providing translation invariance—ensuring the model recognizes a tumor regardless of its exact coordinate in the MRI frame.

And finally the output feature maps are flattened into a 1D vector and passed through two Fully Connected (Dense) layers. The penultimate layer uses a ReLU activation to introduce non-linearity, while the final layer uses Softmax to output a probability distribution across the four tumor classes.

We use ReLU, which stands for **Rectified Linear Unit (ReLU)**, activation function to address the vanishing gradient problem and allow the model to learn complex, non-linear patterns within the medical imagery:

$$f(x) = \max(0, x)$$

The depth of this baseline model was chosen to reflect the "Vanilla" approach taught in class. In the first layer, the filters typically activate in response to simple edges. Then goes to the second layer and third layer, they combine these edges to detect shapes, such as the circularity of a pituitary tumor or the irregular branching of a glioma. Lastly, the final dense layers act as the "reasoning" center, mapping these high-level shapes to the specific categorical labels required by the project objectives.

While the model lacks the advanced residual connections of our later architectures, its inclusion is still critical for a rigorous evaluation. It provides a "control group" that allows us to quantify exactly how much performance gain is attributed to the architectural complexity of the Advanced Model versus the quality of the raw data.

5.2 Advanced Model architecture

While the baseline CNN relied on a standard stack of layers, our advanced approach utilizes **EfficientNet-B0**. This model was selected because it achieves state-of-the-art accuracy with significantly fewer parameters than traditional models like ResNet, making it ideal for medical imaging where precision and computational efficiency must be balanced.

The core innovation of EfficientNet is **Compound Scaling**. In traditional deep learning, models are typically scaled by increasing only depth (layers), width (channels), or image resolution. EfficientNet-B0, however, uses a compound coefficient ϕ to uniformly scale all three dimensions. By scaling all the dimensions it means scaling the mathematical constraints:

- Depth (d): $d = \alpha^\phi$
- Width (w): $w = \beta^\phi$
- Resolution (r): $r = \gamma^\phi$

By optimizing these parameters together, the model ensures that as the network gets deeper to capture

complex tumor patterns, it also gets wider to capture fine-grained textures and uses higher-resolution inputs to maintain spatial clarity.

The fundamental building block of EfficientNet-B0 is the MBConv (Mobile Inverted Bottleneck Convolution). There's a significant difference between this structure and CNN:

1. **Inverted Structure:** Unlike residuals where we squeeze the channels, MBConv first expands the input to a high-dimensional space using a 1×1 expansion layer. This allows the model to learn more complex representations of the brain tissue.
2. **Depthwise Convolutions:** To maintain efficiency, the model performs convolution separately for each channel. This drastically reduces the total number of parameters (3×3 vs. 1×1 operations) without sacrificing the ability to identify tumor boundaries.
3. **Squeeze-and-Excitation (SE) Optimization:** Each block includes an SE optimization step, which explicitly models the interdependencies between channels. This allows the network to focus on the most informative features of the MRI—such as the specific contrast of a lesion—while ignoring background noise.

Moreover, we implement a technique called transfer learning and have two phases to train it. In the first phase, the initial layers of EfficientNet-B0, which detect universal features like edges and blobs, were kept frozen during training to preserve the robust feature extractors learned from millions of general images. We keep the last few layers(classification layers) unfreeze so that the model adapts to our specific task, which is brain tumor classification. We append a custom classification head consisting of a **Global Average Pooling** layer to reduce the feature map followed by a **Dropout layer (p=0.5)** to prevent overfitting. The final Softmax layer was tuned specifically for our four brain tumor categories: Glioma, Meningioma, Pituitary, and No Tumor. We train it for 10 epochs and see that the accuracy is pretty terrible(around 32%) , which is expected because our pretrain model was mostly trained on general images like dogs, cats, or cars.

Then in the second phase we unfreeze every parameter and train it on the much smaller learning rate($1e-5$) so that the model won't completely forget what it has learned but still extract some important information from the brain tumor images. Specifically, we give custom weights to each class so that the model will focus on class what has given more weight:

- 0(glioma): 1.5
- 1(meningioma): 2.2
- 2(No Tumor): 1.0
- 3(Pituitary): 0.6

The reason we do this is because we notice in the initial training, class glioma and meningioma's recall and precision are significantly lower than the other two classes. By giving more weights to these classes, the model will be more sensitive when it's doing classification for them.

Finally, the choice of EfficientNet-B0 over a deeper variant (like B7) was a deliberate decision based on our dataset size. Given that medical datasets are often smaller than general-purpose datasets, using a lighter advanced model like B0 reduces the risk of the model simply memorizing the training images (overfitting) while still providing the sophisticated attention mechanisms needed for high-accuracy diagnosis.

5.3 Optimization and Learning Strategy

For our four-class brain tumor classification task, we utilized **Categorical Cross-Entropy (CCE)** as our

primary objective function. CCE measures the dissimilarity between the true label distribution (y) and the predicted probability distribution ($\hat{y}$) generated by the Softmax output layer.

To update the network weights, we employed the **Adam Optimizer**. Adam is an advanced gradient descent algorithm that computes individual adaptive learning rates for different parameters by estimating the first and second moments of the gradients.

1. **First Moment (m_t):** An exponential moving average of the gradients (momentum), which helps the model navigate through noisy gradients and avoid local minima.
2. **Second Moment (v_t):** An exponential moving average of the squared gradients. It tracks how much the gradients are vibrating or changing. If the changes are huge, Adam slows down and stays stable. If the changes are tiny then it speeds up the gradient.

Since EfficientNet uses **Inverted Residual Blocks**, different layers handle very different levels of abstraction. Adam automatically adjusts to ensure the early edge-detection layers and the late classification layers all learn at their own optimal speeds. Moreover, Adam is generally less sensitive to the initial learning rate than standard Stochastic Gradient Descent (SGD), which allowed us to converge more reliably during the fine-tuning phase of our transfer learning strategy.

To further refine the optimization, we used a **Learning Rate Scheduler**, ReduceLROnPlateau. If the validation loss stagnated for 3 epochs, the learning rate was reduced by a factor of 0.2. This allowed the model to "settle" into the global minimum of the loss landscape once it reached the final stages of convergence. This approach allows the model to travel faster with higher learning rates to find the general region of the global minimum. As the model approaches the minimum, the scheduler "cools" the learning rate, allowing the model to make microscopic adjustments to the weights.

6. Experimental Setup

This part includes the computing environment, preprocessing steps, training procedure, and model configurations. We will show how 2 models were trained and why certain choices were made.

All experiments were operated in a GPU (A100) environment to speed up training. And we implemented the models using Python with TensorFlow and Keras. Since the dataset are MRI images, we first preprocess all images to have the same size : 224 X 224 X 3. This picture size is commonly used in image classification tasks. And it is also required by the EfficientNet architecture. We use the same input size for both models to make our model performance comparison fair, because the difference in performance can come from the input format.

Before the training process, we also normalize the pixels by dividing by 255, so that our input values are in a range of [0,1]. This helps our training process be more stable and allows the optimizer to converge more easily. In addition, we apply several data augmentation techniques, such as rotation, width and height shifts, zooming, and horizontal flipping. These transformations help to collect different types of the same image and add them into the dataset to make it larger . This is important because our dataset is not large enough, and without augmentation our model may easily overfit.

When comparing computation cost, we observe that there is a trade-off between model complexity and efficiency. The baseline CNN is relatively simple, so the number of parameters is less and each training epoch is faster. This guarantees for easy testing and debugging. However, because this baseline model is not deep enough, its ability to capture complex visual patterns is also limited. On the other hand, the

EfficientNet model is more complex. It uses a pretrained backbone and has a larger number of layers. This leads to larger memory usage and longer training time. In addition, the fine-tuning stage requires more computation because more parameters are updated during the training. As a result, the total training time is longer than the baseline model. However, this extra cost is expected, since more complex models usually require more computation but it also has the potential to achieve better performance.

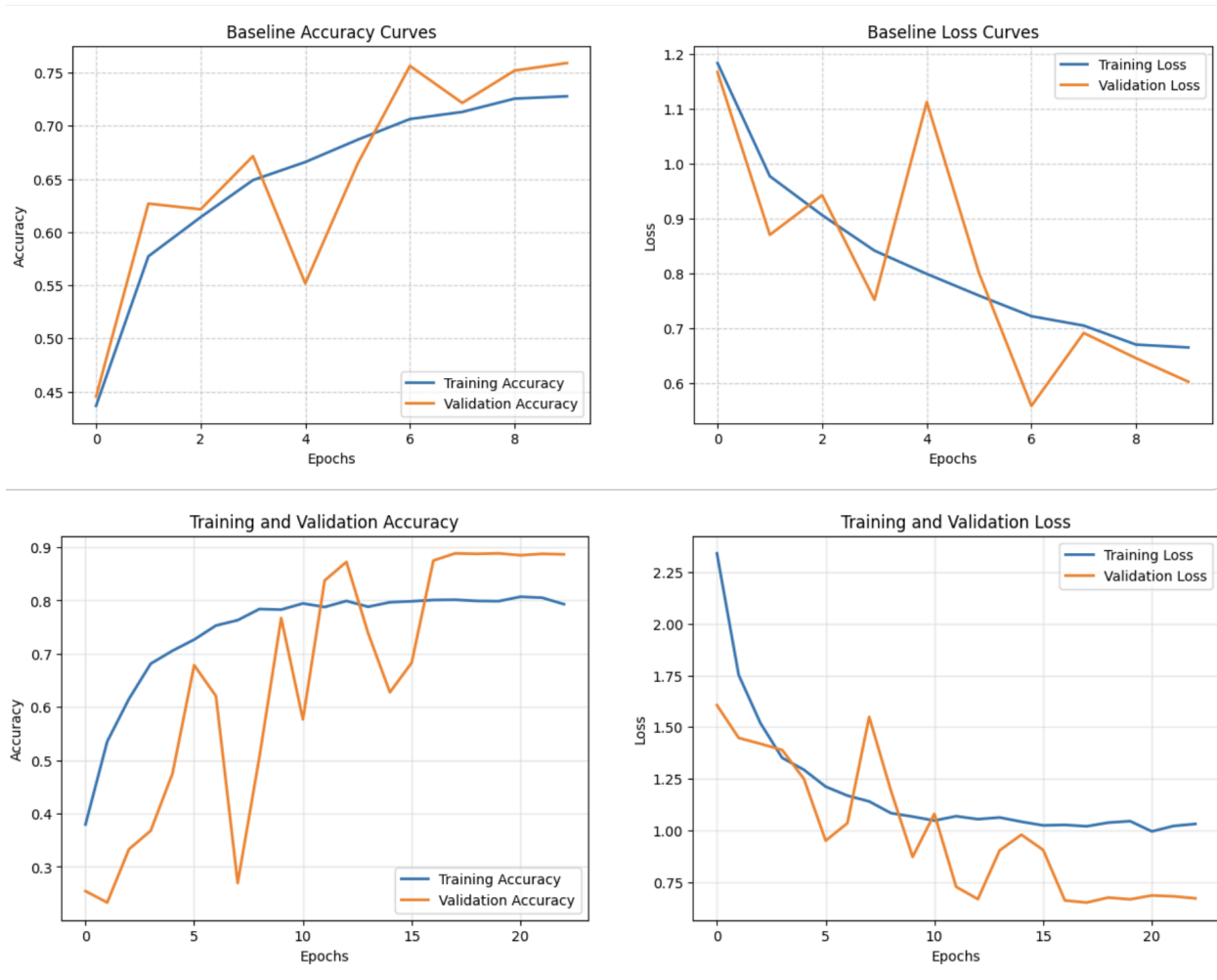
The baseline CNN follows a standard architecture with multiple convolutional , global average pooling layers, and a fully connected classifier. The convolutional layers use small filters to extract local features from the images, and pooling layers reduce the spatial dimensions and also keep then important information. According to our model design , we use the Adam optimizer because it works well in practice and does not need too much manual change. For the batch size, we choose 32 because it provides a balance between training stability and memory usage. We use categorical cross-entropy as our loss function, since we are taking a multi-class classification problem. We also use dropout layers and batch normalization. Dropout randomly removes some neurons during training, which prevents the model overfitting by focusing too much on specific features. And the batch normalization helps the model to stabilize the learning process by keeping intermediate outputs in a reasonable range.

For the advanced model, we use EfficientNetB0 pretrained on ImageNet . The main idea of this transfer learning is that the model has already learned many useful visual features from a large dataset, so that we can reuse these features for our task. Instead of training the model all from EfficientNet, we replace the final classification layer with a new one, which matches our four tumor classes. This new classifier includes global average pooling, batch normalization, dropout, and a dense layer. Global average pooling reduces the number of parameters in our model compared to a fully connected layer, and it helps prevent overfitting.

The training of EfficientNet is divided into two stages. In the first stage, we freeze all pretrained layers and only train the new created classifier. This allows the model to quickly adapt to our task without changing the pretrained features too much. In the second stage, we unfreeze part of the pretrained network and give it a smaller learning rate. This step is important because MRI images are different from natural images in ImageNet. In this case, our model needs to learn these new features to fit better in the new data. We also use early stopping to prevent overfitting, and we also reduce the learning rate when the validation performance stops improving. These techniques help model to make the training process more stable and efficient.

7.Quantitative Results

In this section, we present a detailed comparison between the baseline CNN and the EfficientNet model. We will analyze the results from multiple perspectives : overall performance, class behavior, ROC curves, training and validation curves, and ablation parts. This provides a more complete understanding of how both models perform.



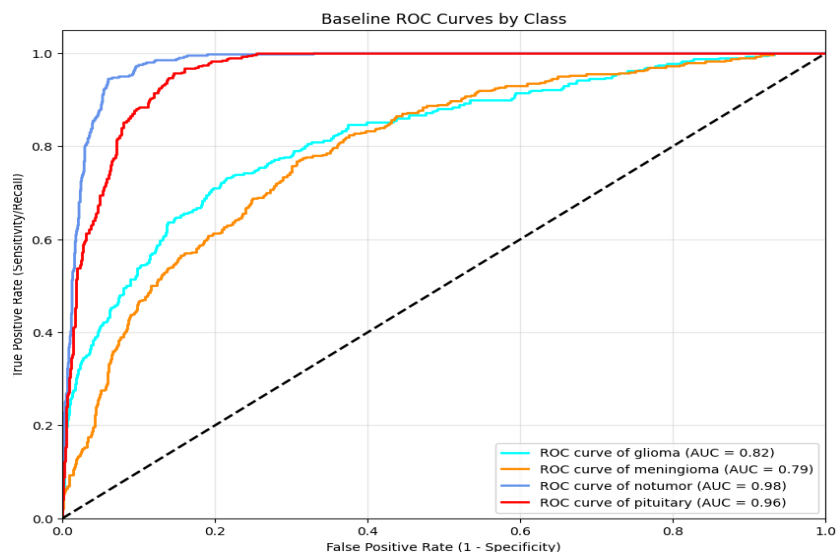
The first set of graphs are for the basic CNN model; the second set of graphs are for EfficientNet.

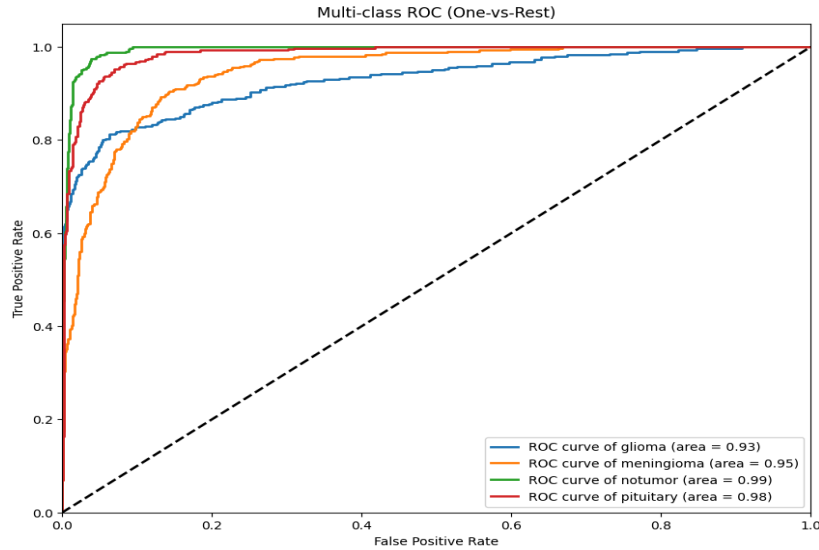
First, we look at the overall performance. The baseline CNN achieves an training accuracy around 73% and validation accuracy about 76%, while the EfficientNet model improves the training accuracy to approximately 80% and validation accuracy about 89%. For the baseline model, the training accuracy increases sharply, and the validation one fluctuates and has a convergence trend in the end. In contrast, the EfficientNet model shows a faster convergence and has a higher validation accuracy. After the fine-tuning, our training process becomes more stable, and the validation loss decreases more smoothly. Also, the gap between training and validation performance is smaller, which will lead to a better generalization of the new data. When we look at the precision, recall, and F1-score, the baseline model has about 0.82 F1 score on the easily predicted tumor classes and about 0.46 score on the difficult tasks. In contrast, the EfficientNet model has an overall average F1-score close to 0.85. (about 0.3 increases on hard classifications and 0.1 increases on easy ones). This suggests that our advanced model is not only more accurate but it also has more balanced improvements.

=====				
BASELINE CLASSIFICATION REPORT				
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	precision	recall	f1-score	support
glioma	0.67	0.52	0.58	400
meningioma	0.60	0.40	0.48	400
notumor	0.72	0.98	0.83	400
pituitary	0.74	0.89	0.81	400
accuracy			0.70	1600
macro avg	0.68	0.70	0.68	1600
weighted avg	0.68	0.70	0.68	1600

	precision	recall	f1-score	support
glioma	0.87	0.75	0.81	400
meningioma	0.80	0.73	0.77	400
notumor	0.88	0.97	0.93	400
pituitary	0.84	0.94	0.88	400
accuracy			0.85	1600
macro avg	0.85	0.85	0.85	1600
weighted avg	0.85	0.85	0.85	1600

The class results show more information into where the improvement comes from. The baseline model performs inadequately on certain classes, especially on meningioma, where the recall is around 0.40. This means that more than half of the actual meningioma cases are confused. Similarly, the recall for glioma is around 0.52, which is also relatively low. These results show that the baseline CNN can make usual mistakes on distinguishing between tumor types that have similar visual patterns. After applying transfer learning, the model performance improves efficiently. The recall for meningioma increases to 0.73, and the F1-score reaches about 0.77. For glioma, recall goes to 0.75 and the F1-score improves to 0.81. The “no tumor” class achieves the best performance in both models, which has an F1-score around 0.93 as the best result. The pituitary class also shows a good performance, with an F1-score close to 0.88. Totally, the EfficientNet model makes more balanced and reliable predictions across all classes.





The ROC curves further prove the difference between the two models. For the baseline CNN, the AUC values vary across classes. Some classes are high, and some classes are low. The AUC for glioma is around 0.82 and for meningioma around 0.79, which are relatively low compared to the other classes. The “no tumor” class reaches about 0.98, and pituitary is around 0.96. This indicates that our baseline model performs well on easily classified classes but has problems on more difficult ones. In contrast, the EfficientNet model achieves consistently high AUC values for all classes. The AUC for glioma increases to around 0.93, meningioma improves to around 0.95, and both “no tumor” and pituitary classes reach close to 0.99 and 0.98. The ROC curves are all closer to the top left corner, which means the model has stronger classification ability.

To better understand which parts in the model contribute most to the performance, we conduct ablation analysis. The results show that data augmentation plays an important role in improving generalization. Without augmentation, our models tend to memorize the training data and perform worse on validation data. Adding transformations such as rotation, shifting, and flipping helps the model learn more robust features. Dropout and learning rate scheduling also improve the stability of training. However, the most important part is fine tuning the pretrained layers. When we only trained the classification head, the performance improvements are limited. After unfreezing part of the network and fine-tuning, the advanced model achieves an obvious increase in both accuracy and F1-score. This shows that using pretrained features to our dataset is useful and important.

Even though the EfficientNet model performs much better at all, it still makes some errors. Most of the confusions occur when different tumor types look visually similar, especially between glioma and meningioma. In addition, low-quality MRI images with low contrast or noise can also have a bad influence on the model’s predictions. These observations indicate that the model still relies on visual features much and may struggle when those features are not clear enough.

In summary, above results illustrate that transfer learning significantly improves the performance of brain tumor classification. The EfficientNet model not only achieves higher accuracy, but also provides more stable predictions and a better generalization ability. At the same time, the analysis also reveals some limitations, that is, we could focus on collecting more data or exploring more advanced architectures in the future.

8. Discussion

8.1 Ablation Studies

To isolate the primary drivers of model performance, we conducted a series of ablation experiments. These studies demonstrate how specific architectural choices and training protocols contributed to the final accuracy.

- **Ablation 1: The Role of Data Augmentation**

We evaluated the impact of removing our data augmentation pipeline, which included 20 rotations, width and height shifts of 0.2, zoom, and horizontal flips. Without these transformations, the model's training accuracy rapidly approached 1.0 while validation accuracy stagnated, confirming that augmentation is the primary defense against overfitting on a dataset of this size. It forced the network to learn invariant geometric features of tumors rather than memorizing orientation-specific pixel intensities

- **Ablation 2: Regularization and Training Stability**

Our experiments showed that omitting Dropout (0.3 in baseline, 0.5 in advanced) and Batch Normalization led to high volatility in the loss curves. The inclusion of ReduceLROnPlateau and Early Stopping was essential for convergence. Specifically, dropping the learning rate by 80% after 3 epochs of no improvement allowed the model to settle into a more stable local minimum during the fine-tuning phase

- **Ablation 3: Fine-Tuning vs. Frozen Backbone**

A comparison was made between Phase 1 (training only the classification head) and Phase 2 (unfreezing the EfficientNetB0 backbone). Phase 2 resulted in a significant performance jump. While the ImageNet-pretrained weights provided a strong initial feature set, unfreezing the layers allowed the deep convolutional filters to adapt to the specific textures and grayscale contrasts unique to medical MRI scans

8.2 Error Analysis and Failure Modes

Despite achieving a final macro average F1-score of 0.85 with the advanced model, certain failure modes persist.

- **Visual Ambiguity (Glioma vs. Meningioma):** The model struggled most with distinguishing between glioma and meningioma, which often share similar morphological characteristics in low-contrast scans. Meningioma recorded the lowest F1-score (0.77) and recall (0.73), frequently being mislabeled as glioma
- **Class-Specific Performance:** The "notumor" class was the most robust, achieving an ROC-AUC of 0.99 and an F1-score of 0.97. This suggests the model is highly effective at screening for the absence of pathology, though less precise at specific differential diagnosis
- **Sample Edge Cases:** Analysis of error images showed that misclassification often occurred in scans with significant noise or artifacts, where tumor boundaries were poorly defined

8.3 Interpretation

The results align with our initial hypothesis: deep learning models, particularly those leveraging transfer

learning, can effectively automate medical classification tasks. EfficientNet's ability to capture complex visual features far exceeds that of a simple CNN. However, the remaining errors suggest that the model lacks the "global" anatomical context a human radiologist utilizes, such as the specific location within the brain where certain tumors are more likely to occur.

9. Conclusion & Future Work

9.1 Summary of Findings

We have partially solved the problem of automated brain tumor classification. Our study demonstrates that an advanced model using EfficientNetB0 significantly outperforms a baseline CNN across all metrics, achieving 85% overall accuracy compared to the baseline's 70%. The inclusion of data augmentation and phased transfer learning proved vital for adapting the model to a specialized medical domain.

9.2 Limitations

The primary limitation remains the dataset size (7,021 total images), which is relatively small for deep learning standards. This limits the model's ability to generalize to rare tumor subtypes or different MRI hardware/protocols.

9.3 Future Work & Academic Extensions

- **Computational Expansion:** With a six-month extension and a tenfold increase in compute, we would implement Vision Transformers (ViT) to capture long-range spatial dependencies that CNNs may miss. We would also expand the training set to include multi-planar views (axial, sagittal, and coronal) for every patient. In addition, we would explore hybrid models that combine CNN feature extraction with transformer-based attention mechanisms, which may further improve performance by leveraging both local and global information.
- **Data Expansion & Generalization:** To improve model robustness, we would incorporate additional datasets from different sources or institutions. This would help reduce dataset bias and improve generalization to unseen data. We would also apply more advanced data augmentation techniques, such as elastic deformation and intensity scaling, to better simulate real-world variations in MRI scans and reduce overfitting.
- **Masters Thesis Standard:** To elevate this to a Master's thesis, we would move from 2D classification to 3D Volumetric Segmentation. This would involve implementing a 3D U-Net to not only identify the tumor type but also segment its volume, which is critical for measuring treatment response. Compared to classification, segmentation provides more detailed and clinically useful information, especially for tracking tumor growth over time.
- **Explainability & Clinical Trust:** Furthermore, we would integrate Explainable AI (XAI) techniques like Grad-CAM to ensure the model's "reasoning" is clinically sound. This would allow us to visualize which regions of the MRI images contribute most to the model's predictions and verify that the model focuses on relevant tumor areas rather than noise or irrelevant patterns.
- **Real-World Deployment:** Finally, we would explore deploying the model in a real-world setting, such as integrating it into a simple clinical decision support system. This would require optimizing model size,

improving inference speed, and ensuring stability across different hardware environments. Evaluating the model on external datasets would also be necessary to confirm its reliability in practical applications.

Contribution statement:

1. George Wu:advanced model building/training(and testing)
2. Jiawen Tang:data preprocessing/EDA/Data pipeline
3. Jingchun Zhang: base model building/training(testing)
4. Zhongyang Fan: final evaluation and metrics analysis

Sources: <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

CodeBase:

https://github.com/georgewuu/stats_426-Final-Project/blob/7de296fec7526d89b03b2dd8c6255bc300fb5331/stats426_project.ipynb

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